Film Dosimetry

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Contents

1 Objective

To establish the sensitometric calibration curve of GAFchromic™ film and utilize it for precision dosimetry in external beam radiotherapy.

2 Apparatus

- GAFchromic™ EBT3 Film
- Teletherapy Unit (Linear Accelerator or Co-60)
- Flatbed Document Scanner (e.g., EPSON 12000 XL)
- Solid water phantoms and localized densitometric analysis software

3 Theory

Radiochromic film has become a cornerstone in advanced radiation dosimetry due to its superior spatial resolution, near-tissue equivalence, and self-developing nature. Unlike traditional silver halide radiographic films, radiochromic films do not require wet chemical processing, thus eliminating darkroom uncertainties [1].

3.1 Physical Principles of GAFchromic Films

Modern GAFchromic™ films (e.g., EBT3) consist of an active layer comprising diacetylene monomers, sandwiched between two matte polyester substrate layers. Upon exposure to ionizing radiation, the monomers undergo a solid-state polymerization reaction. This process alters the molecular structure, absorbing visible light primarily in the red spectrum (around 633 nm), leading to a progressive darkening of the film [2]. The degree of polymerization, and thus the macroscopic optical density (OD), is monotonically related to the absorbed dose.

The net optical density (netOD) is defined as:

$$\text{netOD} = \log_{10}\left(\frac{I_{\text{unexp}}}{I_{\text{exp}}}\right) \tag{1}$$

where I_{unexp} and I_{exp} are the transmitted light intensities of the unexposed and exposed films, respectively.

3.2 Dosimetric System and Readout

The quantification of dose requires a meticulously calibrated flatbed document scanner (e.g., EPSON 12000 XL). During readout, the orientation of the film relative to the scan direction must be kept strictly consistent due to the anisotropic, highly organized polymer chains that cause polarization effects [2].

Furthermore, flatbed scanners typically exhibit a “lateral response artifact,” caused by the variation in the angle of incidence of the scanner’s light source along the array axis. A robust QA protocol minimizes these artifacts through central scanning, multi-channel (RGB) analysis, and maintaining a constant post-irradiation development window (usually 24 hours) to account for asymptotic post-exposure polymerization.

4 Observation

5 Conclusion

A comprehensive understanding of the governing physical principles and stringent Quality Assurance protocols ensures optimal clinical outcomes and fundamental radiation safety.

References

[1] E. B. Podgorsak, *Radiation Oncology Physics: A Handbook for Teachers and Students*. International Atomic Energy Agency, 2005.

[2] S. Devic *et al.*, "Aapm task group 235 report: Radiochromic film dosimetry," *Medical Physics*, vol. 43, no. 10, pp. 5556–5584, 2016.